This afternoon

a · Half a page

what you learned · how it lands in your own research

not a lecture summary

b · Frame your project

your problem, framed · five headings · the worksheet, filled in

You can also choose between one of X hackathons proposed.

Frame your own project

four questions, in this order, before a single line of code

1	→	2	→	3	→	4
the question		the data		the purpose		enough?

AI4ChemicalSciences BootCamp 2026 · hackathon kickoff

Plan

1 · A question worth answering

2 · The data you actually have

3 · What the model is for

4 · Enough data, or not?

5 · The flash talk, and what done looks like

The same interest, twice

Vague idea		Open question, precisely defined
drug design	*“Can ML help my series?”*	Can a model trained on our 340 measured analogues rank the next 40 for hERG better than cLogP alone?
materials	*“I want to use ML for MOFs.”*	Can we predict CO_2 uptake at 1 bar within 0.4 mmol/g for MOFs built from linkers absent from the training set?
organic	*“ML for reaction optimisation.”*	Does a model on our 310 C–N couplings pick a better 24-well plate than our current screening panel?
computational	*“Can ML replace DFT?”*	Can a Δ-model on 2,000 GFN2 geometries reproduce DFT barriers to 1 kcal/mol for our substrate class?

Plan

1 · A question worth answering

2 · The data you actually have

3 · What the model is for

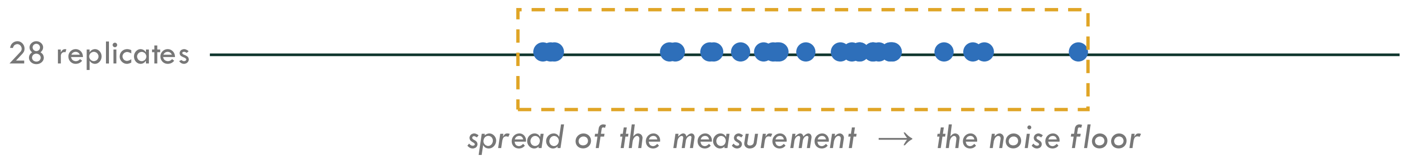
4 · Enough data, or not?

5 · The flash talk, and what done looks like

Data Inventory

source	N (deduplicated)	label + its error	negatives recorded?	cost per new point
our notebooks, 5 people	310 C–N couplings	HPLC yield, ± 5 %	partial — "no product", no conditions	3 h, $40
in-house assay	340 analogues	hERG IC_{50}, ± 0.4 log	yes, inactives kept	1 week, $600
CoRE MOF + our 60	12,000 + 60	GCMC uptake / measured	n/a — continuous	4 h GPU
____________	____________	____________	____________	____________

Quantification of the Noise in your Data?



Plan

1 · A question worth answering

2 · The data you actually have

3 · What the model is for

4 · Enough data, or not?

5 · The flash talk, and what done looks like

Defining Model Function/Objective

The function	The objective	The metric	The users
filter	a shortlist, nothing discarded wrongly	recall @ budget (not RMSE)	you, next week
ranker	an ordering of the top few %	enrichment · precision@k · top-k regret	the person picking the plate
interpolator	the model itself, accurate everywhere	RMSE across the whole domain	everyone, for years
generator	candidate structures	validity · novelty · oracle score	a synthetic chemist
explainer	a mechanism-shaped hypothesis	does the follow-up experiment work?	your next paper

Plan

1 · A question worth answering

2 · The data you actually have

3 · What the model is for

4 · Enough data, or not?

5 · The flash talk, and what done looks like

Can existing data answer my question?

yes

no

Design the split.

What held-out set would mimic the way this model is actually going to be used?

The split IS the scientific claim. Nothing else in the project is as consequential.

Design the acquisition.

What is the cheapest set of new measurements that would make the question answerable?

This is a real project, not a consolation prize — and you have had four lectures on it.

Yes — then test set design is the main question

split	held out	the claim you are making	typical home
random	random 20 %	tomorrow's samples look like today's	fixed library, screening triage
scaffold / cluster	whole chemotypes or clusters	I will be shown new chemotypes	drug design, virtual libraries
time / prospective	everything after a date	I will predict the future of this project	any live campaign
leave-one-group-out	a linker, element, substrate class	I will generalise to a class I never measured	materials, methodology
leave-one-lab-out	a whole instrument or operator	someone else will reproduce this	assay-heavy work

Choose the split before you see a single score. Write it in the brief. Do not change it afterwards.

No — then designing the data is the project

design of experiments

use when

cheap · parallel

what you do

factorial, D-optimal, response surface

Tue — Schmidt

active learning

use when

expensive · you want a model

what you do

query where the model is uncertain

Wed — Schleinitz

Bayesian optimisation

use when

expensive · you want the best point

what you do

surrogate + acquisition, EI / UCB

Tue — Schleinitz

literature extraction

use when

the data exists, just not in a table

what you do

RAG over PDFs and SI + hard validation

Thu W1 — Wahler

…and the fifth option: borrow a pretrained model and fine-tune on the 200 points you have. Often correct. Takes an afternoon.

Whichever you pick, the deliverable is the same: name the first 24 experiments. Not a strategy — the list.

Defining baselines

0	**predict the mean** free	if you cannot beat it, stop and say so
1	**one descriptor + a line** cLogP · Hammett σ · tolerance factor	beat the number a chemist computes in Excel
2	**the human** what your group does right now, quantified	the only baseline that decides adoption
3	**random, at the same budget** for any project that selects or designs	this is the control that exposes memorisation

Wallach, I.; Heifets, A. J. Chem. Inf. Model. 2018, 58, 916–932. DOI: 10.1021/acs.jcim.7b00403

Four briefs that would pass

drug design

question
rank 40 new analogues for hERG

data
340 in-house IC_{50}, ± 0.4 log,
inactives kept

job
ranker

step 4
yes → scaffold split,
hold out 3 chemotypes

baseline
beat cLogP alone

materials

question
CO_2 uptake for MOFs with unseen linkers

data
CoRE MOF GCMC (12k)
+ 60 measured in-house

job
interpolator

step 4
yes → leave-linker-out,
then transfer to the 60

baseline
beat gravimetric surface area

organic

question
pick a better 24-well plate for C–N coupling

data
310 couplings, ± 5 % yield;
failures being recovered

job
ranker

step 4
yes → leave-substrate-class-out

baseline
beat our standard panel

computational

question
Δ-learn DFT barriers from GFN2

data
2,000 GFN2 geometries;
40 DFT barriers so far

job
interpolator

step 4
no → active learning,
200 more DFT points

baseline
beat plain GFN2 barriers

Six ways this goes wrong

✗ **duplicates across the split**	same compound, two names, two salts, two tautomers — deduplicate on InChIKey BEFORE splitting
✗ **tuning on the test set**	every look costs you a little of it; keep a validation set, touch the test set once
✗ **descriptors that leak**	batch, date, plate position, sample ID — if MW is your top SHAP feature, ask why
✗ **extrapolation dressed as prediction**	report where the test points sit relative to training; outside the hull it is a guess with a decimal point
✗ **metric mismatch**	optimising RMSE when the job needed enrichment
✗ **no baseline**	still, somehow, the most common one

Artrith, N. et al. Best practices in machine learning for chemistry. Nat. Chem. 2021, 13, 505–508. DOI: 10.1038/s41557-021-00716-z · two pages, free, read it tonight

Plan

1 · A question worth answering

2 · The data you actually have

3 · What the model is for

4 · Enough data, or not?

5 · Peer review

Next Hackathon Sessions

kickoff	framing	afternoon 1	afternoon 2	afternoon 3	flash talks
this talk	worksheet + one 5-min conversation with a TA	Framing + peer review?	baseline, then your first model	honest evaluation, then slides	5 slides · 5 minutes hard stop

Deliverable Five slides. Five minutes. In this order.

1	**Your initial project: question + plan** verbatim from your worksheet + reviewer comments addressed/accepted	45 s
2	**The data** N · label · error · negatives — one table	45 s
3	**The split, or the first *n* experiments** your actual contribution	90 s
4	**Comparison with baseline** two bars minimum	90 s
5	**Something that you did not expect/future remediation plans** and what three more weeks would buy	30 s

"It did not work, and here is the diagnosis" is a strong talk

Successful Project

tier 1

a brief and a baseline

sharp question, honest data inventory,
baseline measured

tier 2

+ one model on your own split

and a number you are willing
to defend

tier 3

+ the diagnosis

why it worked, or why it did not
— rare in three afternoons